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Chinese Lunar Birth Timing in Singapore: New Concerns for Child Quality Amidst Multicultural Modernity

In line with traditional folk beliefs, many Chinese societies throughout the world (with the exception of China itself) began in the 1970s and 1980s to exhibit birth fluctuations during significant lunar zodiac years—baby booms during the auspicious Year of the Dragon and baby busts during the inauspicious (for daughters) Year of the Tiger. This article explores these two inverse natural experiments in multiethnic Singapore. Lunar birth timing has been manifested more strongly as family sizes have declined, a reversal of modernization theory, yet consistent with enhanced concerns for child quality. The social, assimilative, and gender-related dynamics of these preferences are illustrated through an analysis of the seasonality and birth order distribution of lunar birth fluctuations as well as birth patterns among intermarried Chinese mothers and fathers. The article details how lunar birth fluctuations have been influenced by and have influenced official policies instituted by Singapore's shrinking Chinese majority. None of the assimilative social forces discussed here can be expected to weaken lunar birth timing in the future, although government intervention may inhibit its reoccurrence.

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The notion that parents trade quantity for quality of children over the course of industrial development has been addressed in a variety of research on the family (e.g., Caldwell, 1982; Giddens, 1992; Goode, 1963, Hirschman, 1994; Thornton & Lin, 1994). However, the dynamics of this tradeoff have inspired controversy among disciplinary specialists over the years. Becker's (1960) initial economic formulation held that parent's childbearing decisions in more developed societies are akin to purchases of cars, refrigerators, and other consumer durables. From this perspective, rising incomes should lead to increasing desires for both number and quality of children. Blake's (1968) celebrated critique of that model focused on the issue of norms of child quality. Children are unlike other consumer durables because sociocultural norms require more affluent parents to maintain higher standards of upbringing and schooling, which results in their reducing the size of the family, not increasing it. Becker's reformulation (1981) then offered the more nuanced idea that the value of parent's *time* (and attendant concerns for child quantity and quality) is the key commodity addressable within the framework of consumer choice.

Such debates continue today, not only between proponents of self-professedly economic or cultural approaches to childbearing decisions, but also among adherents within each separate approach, and even among integrationists, for whom this po-

larization neglects commonalities between both approaches (Pollak & Watkins, 1993). For instance, few would disagree about the centrality of two issues in this research: child quality and the value of parent's time. Nevertheless, although economic approaches to these components of childbearing decisions seem fairly straightforward (if potentially limited), just what a cultural approach represents is not entirely clear. Critics of the validity of cultural approaches have at least three reasons for skepticism, which are outlined below.

First, there is the question of reductionism. Rational choice approaches imply that culture and ethnicity are residual influences that are relevant only after other socioeconomic characteristics are considered. For example, if ethnic group A has higher fertility than ethnic group B, we should not assume that the discrepancy between them stems from a greater prevalence of pronatal values among the members of group A. Rather, they may simply be responding rationally to their socioeconomic circumstances (e.g., Lee & Lee, 1959). Of course, the standard rebuttal to reductionist approaches is that culture and ethnicity may be related to socioeconomic circumstances in subtle and inextricable ways (e.g., Handwerker, 1986). Indeed, to those for whom the notion of culture is all-encompassing, multivariate attempts to distill a residual impact of culture seem conceptually flawed.

The second problem, resulting directly from the first, concerns the conceptual strategies of most studies purporting to link culture and childbearing. Such studies typically begin by justifying why some dimension of family formation is of general social consequence (e.g., premarital pregnancy, completed family size, parenting style) and then determine whether cultural differentials along that dimension may be reduced to socioeconomic characteristics. However, such a strategy is not necessarily ideal because skeptics may claim that whatever socioeconomic controls are introduced are somehow insufficient. Instead of focusing on the sufficiency of our controls, a better strategy for demonstrating culture's relevance may be to identify what kinds of practices are less socioeconomically reducible to begin with. Indeed, declines in family sizes in industrial societies have led researchers to identify other childbearing patterns through which cultural preferences may be less clearly reducible to socioeconomic circumstances. These include birth intervals (e.g., Rindfuss & Morgan, 1983; Teachman & Polonko, 1985), birth seasonality (Anderton & Barrett, 1990), and preferences for sons (Park & Cho, 1995). However, as these au-

thors themselves discuss, what might be interpreted as cultural patterns in these childbearing indicators also may be reducible to some extent to socioeconomic circumstances. The first birth interval may be influenced by maternity leave benefits, birth seasonality may vary depending on the proportion of a society engaged in agriculture, and son preferences may result in part from contexts in which sons are more economically valued. Thus, those child-bearing practices that are most unambiguously cultural, almost by definition, should tend to be less connected with socioeconomic concerns, *per se*, than social fashion. For instance, in some related research on the family, Lieberman and Mikelson (1995) identify child naming as a practice as irreducible to socioeconomic circumstances as one can imagine.

Third, there are issues of regularity and historical replicability. In a unique case study, Rindfuss, Reed, and St. John (1979) showed that the fertility of Southern Whites in the United States fell temporarily after the Supreme Court's *Brown* versus the Board of Education ruling in 1954, which outlawed school segregation. They theorized that this sudden fertility dip reflected the ethnic anxieties felt by Southern Whites who felt that their way of life had changed after the ruling. A limitation of this study, however, is that this single historical event cannot be replicated. In addition, it is not obvious, *a priori*, why ethnic tensions should have manifested themselves in temporarily lower White fertility, rather than higher. Similar ambiguities have plagued the literature on minority group status and fertility (Goodkind, 1995). Ideally, theories regarding the directional outcome of ethnic tensions on fertility and family formation should not be assembled after the fact; we should know beforehand how ethnic preferences are expressed and be able to test for them regularly over time.

This article identifies a new phenomenon that bypasses the aforementioned criticisms to a greater extent than previously possible and, in so doing, demonstrates the unambiguous influence of culture on childbearing. The phenomenon concerns preferences in many Chinese societies for timing births in accordance with the traditional lunar zodiac calendar. Specifically, since the 1970s, Chinese parents (with the prominent exception of those in the People's Republic of China) have timed births to coincide with the auspicious Year of the Dragon and avoided births in the Year of the Tiger. These patterns of lunar birth timing are unique to Chinese cultural

groups, manifest themselves in predictable directions, and can be assessed at regular 12-year intervals. Such initial investments in child quality precede all other parenting and schooling decisions subsequently undertaken. Furthermore, these parental valuations of symbolic time manifest themselves in short-term shifts in birth timing that have trivial implications for completed family size (e.g., it is a period effect, rather than an age or cohort effect). As will be shown shortly, lower levels of childbearing in the postindustrial age have rendered the fulfillment of such cultural preferences increasingly cost free and, hence, largely irreducible to parents' own socioeconomic circumstances.

CULTURAL ORIGINS OF LUNAR BIRTH TIMING

Lunar birth timing may well seem exotic to Westerners, but it is by no means a cultural curio confined to some remote island society. Lunar birth fluctuations have been documented among a variety of Confucian societies (e.g., Chinese, Vietnamese, Korean, and Japanese) in Taiwan, Hong Kong, Malaysia, Singapore, Thailand, and the Philippines (Goodkind, 1991, 1992, 1995), Japan (Aso, 1978; Kurosu, 1992), South Korea ("Bad Year for Girls," 1990; Park & Cho, 1995), Vietnam and Australia (Goodkind, 1994) and the U.S. (Goodkind & Xie, 1996; Kaku & Matsumoto, 1975)—societies that include almost 300 million people. If such fluctuations begin in China (more on this momentarily), that number would increase to over 1.5 billion people, well over a quarter of the earth's population. In addition to many other commonalities, Confucian societies all use China's ancient lunar zodiac calendar, which consists of a cycle of 12 animals. Five elements that interlock with the animal cycle (earth, fire, water, wood, and metal) further differentiate these years into a 60-year cycle. The years considered favorable or unfavorable for births vary somewhat among these societies. Parents in Chinese societies tend to favor children, particularly sons, born during the Year of the Dragon (Sun, Lin, & Freedman, 1978) because the dragon is closely associated with health, dynastic nobility, and longevity. Conversely, some Chinese consider the Year of the Tiger an inauspicious time to bear a daughter because that animal is associated with ferocious, unfeminine characteristics (Saw, 1975).

Despite the long history of this folk symbolism, lunar birth fluctuations among Chinese societies actually did not begin until the 1970s. The

belated blooming of this cultural tradition can be explained in part by a lack of reversible modern contraceptives prior to the 1970s (Goodkind, 1991; Sun et al., 1978). However, contraceptive ability was neither a necessary nor sufficient condition for zodiacal birth timing. In Taiwan, for instance, about half of the 1976 Dragon Year baby boom was due to noncontraceptive strategies of the parents, strategies that entailed marriage timing, abortion, and coital frequency (Goodkind, 1993). Similarly, in a phenomenon analogous to the Tiger Year, Japanese have avoided births in the inauspicious (for girls) Year of the Fire Horse since the 18th century, long before modern methods of contraception were available (Aso, 1978; Kurosu, 1992).

Thus, to explain lunar birth timing phenomena, we need to consider not only the family formation strategies available to parents but also the sociohistorical forces that motivated the use of these strategies. Previous research has suggested the pivotal role of ethnic identification and institutional context. For instance, the aforementioned absence of lunar birth fluctuations in China was due not only to China's one-child policy but also to the social upheavals that occurred during and after the Cultural Revolution (1966-1976) when citizens were exhorted by the state to condemn all ideas deemed to be "superstitious." In fact, China's destruction of traditional symbolism contributed to a converse glorification of such symbols among Chinese societies abroad, which in turn helped to jump-start this new tradition (Goodkind, 1991). Similarly, within the districts of peninsular Malaysia, Chinese were more likely to engage in lunar birth timing to the extent that they constituted a smaller minority (Goodkind, 1995), a finding that suggests that minority status enhanced, rather than diluted, this manifestation of ethnic identity.

However, the social dynamics through which these preferences in family formation may be perpetuated, altered, or dissolved away in the postindustrial age are still not well understood. This article progresses beyond previous research by exploring the interplay of development, gender, ethnic assimilation, and state policy in motivating these phenomena in multiethnic Singapore, where Chinese lunar birth swings have been the sharpest in the world. It begins by examining whether birth fluctuations in the Dragon and Tiger years have, indeed, been enhanced as family sizes have declined. The social dynamics of these decisions are illuminated by seasonal and birth order distri-

butions of fertility changes in relevant zodiac years. Next, birth trends among Chinese mothers and fathers who have married outside their ethnic group are examined to determine if such intermarried parents were less likely to demonstrate lunar preferences and if the gender of the Chinese parent played any role in these negotiations. Lastly, the article shows how decisions about lunar birth timing were affected by (and in turn affected) Singapore's family and ethnic policies.

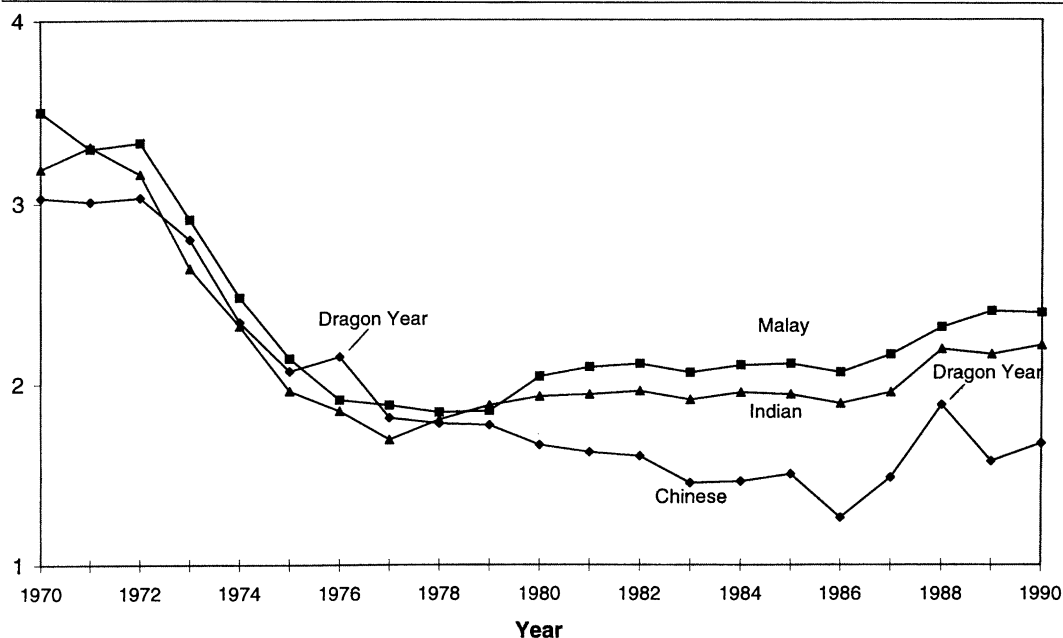
SOURCES OF DATA AND METHODS OF ANALYSIS

The primary evidence used to address these issues is derived from Singapore's Vital Registration, a high quality source of data well suited for an analysis of fertility changes. Most methods of analysis entail basic computations of fertility changes among Singapore's ethnic groups, with comparative inferences drawn from the results. The author's interpretation of these data were supplemented by interviews with population specialists and other informants in Singapore (conducted in English and Mandarin) during a two-week visit in late 1990. That visit was part of a larger field study of this phenomenon in Asia and the West that has continued to the present.

SINGAPORE: LUNAR BIRTH FLUCTUATIONS IN A MULTICULTURAL CITY-STATE

Singapore is a prosperous city-state of 2.5 million people located at the southern tip of peninsular Malaysia. Its current gross national product per capita of \$20,000 is the second highest in Asia, after Japan. Literacy rates are high, and fertility is quite low among all its primary ethnic groups—Chinese, Malays, and Indians. Singapore's unusually low levels of childbearing can be attributed to a trio of factors: a high level of urbanization, substantial economic development, and an antinatalist program regarded as one of the most comprehensive in the world during the 1970s (Saw, 1990). Singapore's official program of national family planning began in 1966 although the full range of antinatalist measures were not legislated until the early 1970s when the government attempted to create a smaller norm of family size. The slogan of this campaign was "two is enough," and Figure 1 suggests that this goal was successfully reached. The total fertility rate, which represents the expected number of births per woman's reproductive life, declined among all ethnic groups from over 4 in 1965 (not shown), to over 3 in 1970, to 2.1 or fewer by 1975. The relative population proportions among the three major ethnic groups remained fairly

FIGURE 1. TOTAL FERTILITY RATES (EXPECTED BIRTHS PER WOMAN'S LIFETIME) AMONG SINGAPORE'S THREE MAJOR ETHNIC GROUPS, 1970–1990



steady from the 1960s to the early 1980s, with the majority Chinese comprising about 78% of the population, Malays about 14% percent, and Indians most of the remaining 8%. By the late 1980s, however, Chinese fertility had fallen to exceptionally low levels (unlike that of Malays and Indians), the causes and consequences of which will be addressed shortly.

Figure 1 indicates that, amidst this context of rapidly declining fertility, only Chinese fertility has been affected by the lunar calendar. Chinese fertility fell notably during the inauspicious Tiger Years of 1974 (Saw, 1975) and 1986 and rose during the auspicious Dragon Years of 1976 and 1988. These fluctuations stand in sharp contrast to the relatively steady fertility among Malays and Indians.

Table 1 shows the size of fertility fluctuations for each ethnic group in lunar cycles since the 1930s. The amplitude of the swings is measured as the proportionate difference between the crude birth rate (births/population) in each zodiac year compared with the average rate of adjacent years. For instance, in 1939 the Chinese crude birth rate was 50.5. During 1938, the Chinese Tiger Year crude birth rate was 3.8% lower than the average of 1937 and 1939, and during 1940, the Chinese Dragon Year crude birth rate was 2.7% lower than the average of 1939 and 1941. Note that our

hypothesized lunar fertility swings did not occur until the 1970s. Prior to the 1970s, Chinese fertility did tend to fall in Tiger Years, but the decline was not very large and was sometimes matched by declines among other ethnic groups. Similarly, prior to the 1970s, Dragon Year fertility did not spike upwards. However, as fertility fell to lower levels in the 1970s, strong zodiacal fertility fluctuations emerged. Tiger Year fertility dropped by 2.3% in 1974 and by 14.3% in 1986, and Dragon Year fertility rose by 11.8% in 1976 and by a startling 22.8% in 1988. Similar fluctuations did not occur among other ethnic groups. The increased amplitude of zodiacal birth swings over time makes sense, assuming that the prevalence of Chinese tastes for such timing remained relatively constant. At progressively lower levels of childbearing, contraceptive use is more common, and parental concerns over child quality become more important.

The timing of marriage also can play a significant role in lunar birth timing, as demonstrated indirectly in Table 2, which shows fluctuations in birth numbers in the Tiger and Dragon Years by birth order. Tiger Year fertility in 1974 and 1986 fell by over 7.5% for all birth orders except the first, which actually rose by 10.7% in 1974. The unusual rise in first births was attributable in part to the substantial increase of marriage in

TABLE 1. AMPLITUDES OF CRUDE BIRTH RATE FLUCTUATIONS DURING TIGER AND DRAGON YEARS (BY PERCENTAGE) AND CRUDE BIRTH RATE LEVELS BY ETHNIC GROUP IN SINGAPORE, 1937–1989

Zodiac Year and Crude Birth Rate	Chinese	Malay	Indian	Others	All Singapore
Tiger Year 1938	-3.8	-1.7	-9.9	-3.9	-3.9
Dragon Year 1940	-2.7	-3.3	2.8	-2.0	-2.5
Crude birth rate 1939	50.5	47.7	29.4	27.6	47.6
Tiger Year 1950	-1.9	-1.4	1.1	-0.9	-1.5
Dragon Year 1952	-0.5	2.7	0.3	-0.2	0.0
Crude birth rate 1951	45.1	45.0	42.7	51.7	45.0
Tiger Year 1962	-1.7	-0.3	-1.3	1.9	-1.5
Dragon Year 1964	0.2	3.7	1.8	-5.5	0.8
Crude birth rate 1963	30.8	45.0	35.3	32.3	33.2
Tiger Year 1974	-2.3	-1.4	1.1	1.3	-1.9
Dragon Year 1976	11.8	0.9	2.1	4.6	9.3
Crude birth rate 1975	17.7	16.9	15.6	29.8	17.7
Tiger Year 1986	-14.3	-2.9	-3.4	-2.9	-10.8
Dragon Year 1988	22.8	1.1	2.1	-2.0	16.0
Crude birth rate 1987	15.0	21.9	20.3	27.7	16.7

Note: Chinese folk beliefs suggest that the Tiger Year is an inauspicious year to have a daughter and that the Dragon Year is an auspicious time to have a son. The crude birth rate is the number of births divided by the population. Amplitudes shown represent the proportional difference in the crude birth rate of each year compared with the average of the two adjacent years. (The low crude birth rate of Indians and others around 1939 reflects the high proportion of male immigrants among these ethnic groups.) The total fertility rate (used in Figure 1) more accurately depicts family sizes in each period, but is not available for "others" and all years before the 1950s. Data are derived from Singapore's Vital Registration, 1937–1989.

TABLE 2. AMPLITUDES OF CHINESE FERTILITY CHANGES (BY PERCENTAGE) IN SINGAPORE DURING TIGER AND DRAGON YEARS BY BIRTH ORDER OF CHILD

	Birth Order of Child						
	All	1	2	3	4	5	6+
Tiger Year 1974	-2.3	10.7	-8.5	-9.4	-7.6	-8.7	-17.8
Tiger Year 1986	-14.3	-1.7	-8.3	-13.8	-19.3	-12.5	-15.5
Dragon Year 1976	11.8	14.3	11.6	12.6	6.1	0.2	-10.2
Dragon Year 1988	22.8	16.9	24.4	33.7	21.3	11.7	-15.1

Note: These amplitudes represent the proportional difference between births in each zodiac year compared with the average number of births in the 2 years adjacent to it. Data are drawn from Singapore's Vital Registration, 1973-1989.

1973-1974 (not shown). Table 2 further exposes some demographic differentials in the manifestation of these tastes. The decline in Tiger Year births was shared among women at all birth orders after the first, but increases in Dragon Year births were weaker among women having fifth or higher order births. The weakness of upward Dragon Year fertility fluctuations among women with many children likely reflects either their lack of control over fertility or simply the swift decline of their desires regarding family size across this era.

Unfortunately, Singapore's published vital registration did not begin to disaggregate births by both ethnicity and education until 1988, so the relationship between education and these reproductive tastes cannot readily be deduced. In other areas where the phenomenon has occurred, rising levels of education do not appear to diminish this behavior. In Taiwan, for instance, an individual-level multivariate model controlling for contraceptive use indicated that lunar birth timing was widespread throughout the literate social classes, although strongest among the middle class (Goodkind, 1992). A district-level analysis in Malaysia similarly suggested wide social prevalence that transcended educational attainment (Goodkind, 1995).

Table 3 further illuminates the strategic scenario facing couples wishing to time births to fall into (or outside of) a significant zodiac year. It depicts the size of seasonal fertility fluctuations measured by the lunar calendar, that is February to January (the Chinese lunar new year falls between late January and mid-February). Table 3 divides the lunar year into quarters and compares fertility in each lunar quarter with the same lunar quarter of the year prior to it. The seasonal patterns differed strongly between the inauspicious Tiger Years and the auspicious Dragon Years. Fertility declines during the Tiger Years of 1974 and 1986 were spread out fairly evenly across the

lunar quarters of each year (e.g., in 1974: -12.6%, -13.0%, -13.0%, -18.6%; in 1986: -12.1%, -12.1%, -13.6% and -20.6%). The somewhat greater decline in the last lunar quarter was an artifact of an upswing in births in the last lunar quarter of the baseline Ox Year by parents wishing to avoid the inauspicious Tiger Year (Goodkind, 1992). This steady seasonal pattern of decline likely results from antinatal strategies that rely primarily on contraception and abortion throughout the conception interval, which would stretch from approximately May of the prior Ox Year to April of the Tiger Year. Indeed, an identical seasonal pattern of birth decreases also occurred during the similarly inauspicious 1966 Year of the Fire Horse in Japan, where changes in contraception and abortion accounted for over 90% of the temporary drop in fertility that year (Aso, 1978).

Changes in the use of contraception and abortion also have contributed to birth rises during the auspicious Dragon Year, but only about 65% of the total (Goodkind, 1993). The rest was due to changes in the volume and timing of marriage as well as the frequency of coitus. These proceptive factors (Miller, 1987) explain why there was a bunching of birth increases in the last two quarters of the Dragon Year (e.g., in 1976: 0.3%, 7.4%, 17.8%, 8.1%; in 1988: 26.2%, 16.1%, 37.5%, 28.6%). The conceptions of Tiger babies could be avoided throughout the entire conception interval, but some months could be required before a Dragon baby was successfully conceived. In addition, as time ran out of the 1-year conception interval, couples would have progressively fewer months to conceive a child, which would increase the necessity of frequent coitus during the latter part of the interval (Goodkind, 1993). Coincidentally, the third quarter of the conception interval for Dragon babies (from November to January) overlaps with the holiday season around the Chinese lunar new year. During that time, po-

TABLE 3. SEASONAL PATTERN OF LUNAR CRUDE BIRTH RATES AMONG CHINESE IN SINGAPORE

Zodiac Animal	Year	Lunar Crude Birth Rate (per 1000)	Annual Change in Crude Birth Rate (by Percentage)	Change in Quarterly Lunar Crude Birth Rate from Same Quarter of Prior Year (by Percentage)			
				Feb–Apr	Mar–Jul	Aug–Oct	Nov–Jan
Ox	1961	33.2	—	—	—	—	—
Tiger	1962	31.4	–5.5	–6.0	–6.3	–3.8	–5.9
Rabbit	1963	30.9	–1.5	–2.7	–2.1	1.1	–2.3
Dragon	1964	28.9	–6.6	–2.6	–7.2	–8.9	–7.0
Snake	1965	27.3	–5.4	–8.0	–4.6	–3.3	–5.8
Ox	1973	22.1	—	—	—	—	—
Tiger	1974	19.0	–14.1	–12.6	–13.0	–13.0	–18.6
Rabbit	1975	17.7	–6.8	–5.3	–7.3	–12.5	0.7
Dragon	1976	19.2	8.2	0.3	7.4	17.8	8.1
Snake	1977	16.3	–14.8	–14.0	–16.6	–16.8	–13.4
Ox	1985	15.1	—	—	—	—	—
Tiger	1986	12.7	–15.5	–12.1	–12.1	–13.6	–20.6
Rabbit	1987	15.3	20.9	7.1	23.3	18.9	39.0
Dragon	1988	19.3	25.9	26.2	16.1	37.5	28.6
Snake	1989	15.7	–18.4	–7.0	–15.1	–24.9	–19.1

Note: The lunar crude birth rate is the number of births between February and January per 1,000 population. Note that this table measures year-to-year fluctuations, in contrast to Tables 1, 2, and 4. In the case of seasonality, these annual comparisons tend to be more readily interpreted than 3-year comparisons. Data are drawn from Singapore’s Vital Registration, 1960–1989.

tential parents are further reminded of the upcoming zodiac year by a plethora of animal symbolism appearing on almanacs, billboards, television commercials, and newspapers.

One may wonder whether lunar birth preferences have, indeed, been sex-specific. For instance, as previously indicated, traditional folklore suggests that parents might wish to avoid having daughters, not sons, in the Tiger Year. How might parents achieve both lunar and sex preferences simultaneously? One potential way could be for parents to misreport their daughters’ birthdays. A second possibility concerns the sex-selective abortion of female fetuses. Indeed, in Singapore, as in other Confucian societies, son preferences and advances in early prenatal sex-testing technologies have contributed to a rising sex ratio (male/female) at birth (Goodkind, 1996). However, annual sex ratios at birth in Singapore and in many other Confucian societies (not shown) provide, at best, inconsistent evidence of interactions between lunar and gender preferences, a finding that may be due to selection effects. That is, risk-averse Chinese parents who were worried about bearing a daughter in the Tiger Year simply may have avoided conceiving altogether. If so, those left to make decisions about misreporting birthdays or sex-selective abortion would be self-selected to be less concerned about lunar folklore, in general.

CHINESE INTERMARRIAGE AND ASSIMILATION:
DAMPENING LUNAR TASTES?

The above findings indicate that developmental forces have not caused Chinese reproductive preferences to converge with those of other ethnic groups. On the contrary, progressively smaller family sizes (and the attendant greater control of fertility through contraception) have enhanced the manifestation of lunar birth fluctuations. But should other assimilative forces be expected to weaken this phenomenon in the future? Intermarriage with other ethnic groups, for instance, is often regarded as the ultimate effect (as well as a future cause) of structural assimilation in plural societies (e.g., Gordon, 1964). To what extent do Chinese Singaporeans intermarry, and are lunar birth fluctuations diluted among such couples? Some weakening might be anticipated, given that this practice requires the cooperation of both spouses, and only the Chinese member of an intermarried couple would presumably care about lunar timing. We might hypothesize further that, to the extent that the husband’s preferences tend to dominate reproductive outcomes, intermarried Chinese fathers would manifest stronger lunar birth fluctuations than intermarried Chinese mothers. Singapore’s vital statistics are well suited to addressing the above hypotheses because annual marriage and birth data both indicate the ethnicity of each parent.

TABLE 4. AMPLITUDES OF FERTILITY FLUCTUATIONS DURING TIGER AND DRAGON YEARS BY ETHNIC GROUP OF MOTHER AND FATHER FOR SINGAPORE, 1973–89

Ethnic Group of		Tiger 1974	Dragon 1976	Number of Births 1975	Tiger 1986	Dragon 1988	Number of Births 1987
Mother	Father						
Malay	Malay	–1.5%	–0.5%	5284	–2.9%	0.4%	8063
Indian	Indian	1.3%	1.5%	1917	–3.4%	2.5%	2673
Chinese	Chinese	–2.5%	12.1%	30438	–14.5%	23.2%	29369
Chinese	All non-Chinese	8.2%	–0.8%	552	–6.8%	7.9%	774
All non-Chinese	Chinese	–7.6%	3.9%	197	–3.8%	12.8%	470

Note: These amplitudes represent the proportional difference between births in each zodiac year compared with the average number of births in the 2 years adjacent to it. Despite annual births in hundreds, amplitude differences between intermarried Chinese mothers and intermarried Chinese fathers are significantly different at the $p < .05$ level in each zodiac year. (Statistical tests of the differences between two proportions are not shown.) Data are drawn from Singapore's Vital Registration, 1973–1989.

According to marriage data, Chinese intermarriage remains rare (Kuo & Hassan, 1976; Lee, Potvin, & Verdieck, 1974). Between the late 1960s and the mid-1980s, intermarriage rates for Chinese men increased from only about 1.0% to 1.5% and for Chinese women from 2.0% to 2.5% (Lee, 1988). Table 4 indicates the amplitude of changes in birth numbers among this small minority of intermarried Chinese in the significant zodiac years compared with birth averages of adjacent years. Compared with homogamous Chinese couples, intermarried Chinese exhibited smaller fluctuations, a finding consistent with the hypothesized dilution of such preferences. On the other hand, such preferences were not extinguished; intermarried Chinese were more likely to evince these preferences than couples without any Chinese member. Moreover, the amplitudes of zodiacal fertility fluctuations among intermarried Chinese increased sharply between the 1970s and 1980s. This finding was likely due not only to the falling size of the Chinese family but also to declining selection effects among those willing to intermarry. That is, the initial cohort of people willing to intermarry tend to be most marginal to their own ethnic group. As intermarriage rates increase, the characteristics of intermarriers tend to become less marginal to that of their own ethnic group (Blau, 1977).

Further insights can be derived by comparing birth patterns of intermarried Chinese husbands with birth patterns of intermarried Chinese wives. During the 1970s, intermarried Chinese husbands were more likely to show the expected lunar fluctuations in the Tiger and Dragon years (–7.6%, 3.9%) than intermarried Chinese wives (8.2%, –0.8%). Intermarried Chinese wives experienced higher fertility during the 1974 Tiger Year, an un-

expected finding likely due to sharply fluctuating intermarriage patterns in prior years. That finding confirms my initial hypothesis that husbands' lunar preferences (or lack of them) tend to predominate in fertility outcomes. However, half of that pattern changed during the 1980s. Although the preferences of intermarried Chinese husbands, compared with the preferences of intermarried Chinese wives, continued to dominate in the auspicious Dragon Year (12.8% vs. 7.9%), the preferences of intermarried wives dominated in the inauspicious Tiger Year (–6.8% vs. –3.8%).

This gender disparity between the two birth fluctuations in the 1980s is likely attributable to the interplay of three factors. First, there is a decline in selectivity effects among the intermarried. That is, as intermarriage for Chinese women gradually becomes more common, those who do intermarry are increasingly less marginal to their own ethnic group. Second, women may have relatively greater control of short-term antinatal strategies (e.g., contraception and abortion) than they do for short-term pronatal strategies (e.g., intercourse). Third, women and men simply may respond differently to these zodiac symbols. Women may be more worried about the negative folklore concerning female Tiger babies, whereas men may be more affected by the heroic folklore regarding male Dragon babies. This second hypothesis seems plausible based on sex-specific folklore as well as the author's field interviews regarding such preferences in Singapore and elsewhere. Whichever hypothesis is more correct, we can test for this gender disparity between fertility fluctuations in the Tiger and Dragon years after the next zodiac cycle is completed in the year 2000. In fact, future tests may also be performed in the U.S., where vital registration has included

birth statistics for intermarried persons, including Chinese, since 1989.

SINGAPORE'S FAMILY POLICIES, ETHNIC RELATIONS, AND LUNAR TIMING

Of course, lunar birth timing does not simply result from the characteristics of individual parents and their intimate negotiations. Changes in Singapore's family policies have shaped the climate in which the birth-timing decisions of all couples have been affected. In 1987, fertility policy switched from antinatal to pronatal, due in part to concerns about exceptionally low Chinese fertility, which presaged a shrinking political majority. Lunar birth fluctuations affected the timing of this policy shift, which, in turn, had repercussions for subsequent birth fluctuations.

Originally, official reaction to the Dragon Year baby boom in 1976 was quite positive. In fact, Singapore's president, Lee Kuan Yu, congratulated the parents in his New Year's address in 1977 and cast the phenomenon under the rubric of both multiculturalism and nationalism. Lee said, "I wish them [Dragon parents] well. I myself have a Dragon son. . . . No amount of work by the Ministry of Culture or Education could change these beliefs. Many years of experience have taught me that on crucial emotional issues of language, culture, and religions, the *Singaporean* [italics added] . . . has deep abiding values, faiths, myths, and mythology" ("Lee's New Year Message," 1977). Such a statement was congruent with government policies intending to foster a Singaporean identity, a concept without strong historical roots in a society characterized by sharp ethnic cleavages (Bedlington, 1978; Heidt, 1987).

Singapore's government hoped that fertility would remain at replacement levels (Saw, 1990). By the early 1980s, however, Chinese fertility had fallen to levels well below that of other ethnic groups (see Figure 1), due in part to the higher level of education among Chinese. The proportion of Chinese births among all births fell from 77% in the 1970s to 69% in the mid-1980s. If that downward trend continued, the sizeable majority of Chinese in Singapore would be gradually diluted, a prospect that has always frightened Singapore's Chinese leaders. Indeed, Chinese-dominated Singapore, which had gained independence from Britain in 1959 and tentatively joined the Malay Federation shortly afterwards, pulled out of that federation in 1965 for fear of becoming politically dominated by ethnic Malays, who con-

stituted about two thirds of the population of Malaysia (Bedlington, 1978). In response to the perceived crisis of low Chinese fertility in the 1980s, Singapore experimented in the 1980s with a eugenics law intended to increase the proportion of well-educated women who had births (Palen, 1988). Because the best educated were more likely to be Chinese, such a policy was intended to perpetuate a strong Chinese majority.

It was not until 1986, however, that a major reversal in the old two-is-enough policy officially was enacted. That was the Year of the Tiger when Chinese total fertility fell to an unprecedented low rate of 1.2 children per woman. In August of 1986, Second Prime Minister Goh Chok Tong (the current prime minister) announced Singapore's "Second Long March" to reverse its fertility policy. He noted the social problems that would result from overshooting the targets for fertility reduction, such as a future lack of soldiers for national defense. Although pressures to switch from antinatal to pronatal policies were building anyway, the Tiger Year baby bust was a major factor that precipitated it ("Fewer Babies Born Last Year," 1987). A sweeping set of reforms in family policy were announced on March 1, 1987, coincidentally right after the tail end of the Tiger Year baby bust. The old slogan was replaced with a new slogan: "At least two, better three, four if you can afford it." Higher fees for late birth-order deliveries were repealed, sterilization bonuses were eliminated, and new provisions encouraged large families, especially among older and better educated women. These included special tax incentives, extended maternity leave, and special housing allowances and childcare subsidies (Palen, 1988).

Public pronouncements attempted to bolster these pronatal efforts by reminding parents that upcoming lunar years in the zodiac calendar were auspicious for births. For instance, after the Chinese total fertility rate had plummeted to 1.2 births per woman during 1986, President Lee remarked with relief that "we have left the Tiger Year behind us. Only 26,000 Chinese babies were born in the Tiger Year, 15,000 short of 41,000, the replacement level. . . . I hope we can do better this Year of the Rabbit" ("PM Hopes For More Babies in Rabbit Year," 1987). Later on that year, the trade and industry minister indicated some progress in increasing fertility by noting that the Year of the Rabbit and the next year, the Year of the Dragon, were auspicious, but the zodiac could not be depended on to solve the birth

dearth forever ("We Must Stop Danger Trend," 1987). Around the same time, other headlines heralded the impending baby boom of 1988 ("Maternity Hospitals Packed," 1987).

But as the conception interval for auspicious Dragon babies passed in May of 1988, officials realized that the inauspicious Snake Year of 1989 might cause fertility to decline, contrary to pronatal desires. Government leaders now reversed course. "We should not decide the birth of our children by the animal year. Have your baby any year, including the Snake Year," they suggested ("Forget the Zodiac Sign," 1989; "Dragon Year Births Highest," 1989). Such governmental exhortations did not, however, prevent fertility from sliding back sharply during the Snake Year of 1989.

An unanticipated social consequence of lunar birth timing is the staggered size of successive, annual school cohorts. As was the case in Taiwan after 1976 (Goodkind, 1991), the strength of this phenomenon in Singapore in 1988 created headaches for the board of education, which had to assign extra teachers and classroom space to accommodate the 1988 Dragon cohort. A total of nine additional elementary schools accommodating about 4,000 students had to be built for this cohort, although the number of births in the 1988 cohort actually exceeded that of its predecessor by about 9,000 ("Weekly Edition," 1994). To avoid this problem in the future, the government now plans to discourage lunar birth timing of all kinds. An anti-lunar birth timing campaign likely will be enacted before the next Tiger-Dragon cycle begins in 1998 (Saw, 1990).

CONCLUSIONS

Specialists of Western patterns of family formation may view Chinese lunar birth timing as an exotic curio. Such curios, however, sometimes embody natural experimental properties through which grander questions that were previously unanswerable (or even unthinkable) can be addressed. This article has employed the experimental properties of lunar birth timing to draw together in a unique way two central issues in family formation research—parental valuations of time and child quality. Among Chinese Singaporeans, these parental valuations are manifested first by a willingness to speed up or delay childbearing in accordance with the symbolic value they attach to annual blocks of time within the 12-year animal zodiac calendar. Socioeconomic development has contributed to such low levels of childbearing

that, ironically, zodiacal preference patterns have largely swung free from their former socioeconomic moorings. Because this birth timing phenomenon need not stem from parents' current socioeconomic circumstances, it is experimentally valuable. If it is meaningful to search for a baseline of genuinely cultural childbearing preferences in the postindustrial age, this phenomenon will be among the most fruitful for delineating what that baseline is and whether social and assimilative forces may influence it.

The Singaporean case suggests that assimilative forces, which may well reduce social differentials in other aspects of family formation, have so far not weakened this baseline of cultural childbearing preferences. Continued social development and the attendant smaller sizes of families actually have enhanced lunar fertility fluctuations. Ethnic intermarriage should not weaken this phenomenon much—Chinese do not often marry outside their own ethnic group, and even if such marriages become more prevalent, evidence here suggests that lunar preferences among these couples will not disappear. Lastly, although Singapore ostensibly has tried to create a Singaporean identity among its multicultural inhabitants, pro-Chinese family policies have both belied a concern for ethnic separatism and contributed to such separatism. To the extent that future policies contribute to a sharpened sense of ethnic identification, Chinese may well express that identity in their reproductive behavior. The only factor that seems ready to weaken future fluctuations in lunar birth rates in Singapore is a direct government-led attack on zodiac animal symbolism.

These findings also illuminate how parents may bargain over decisions of birth timing. Although evidence from the 1970s suggested that the ethnicity of intermarried Chinese fathers determined birth fluctuations in both the Tiger and Dragon years, in the 1980s, the preferences of intermarried Chinese mothers seemed to dominate in the Tiger Year. The latter finding suggests that men and women either responded differently to these zodiac symbols or they did not control pronatal and antinatal decisions to the same extent. If the latter is a factor, these findings may have implications for other Western, developed societies as well. If mothers' and fathers' preferences for birth timing differ, whose preferences will predominate? An unexpected implication of this study, which will require further testing before it can be confirmed, is that mothers' timing preferences may predominate more when their

concern is to delay a birth than when they wish to have it sooner. (For a broader perspective on couple characteristics and birth timing, see Corijn, Liefbroer, & Gierveld, 1996.) The foregoing study also has contributed to our general understanding of the differing shape of seasonal birth changes that should result when such short-term birth timing intentions are pronatal rather than antinatal.

Have modernizing forces in other societies also allowed cultural beliefs to manifest themselves in new child-timing practices? Although examples as paradigmatic as lunar birth timing are rare, this article has identified criteria by which to search for them in the future. Underlying beliefs about the value of time must be strong, and their associated practices must be both cost-free and attainable. Thus, for instance, it is unlikely that the 12-month Western zodiac (e.g., Aries, Taurus, etc.) will ever affect fertility in the U.S. Although most Americans know their zodiac sign and where to locate their daily horoscope, they are probably less aware of the calendrical procession of these symbols and their associated personality outcomes than Chinese are aware of the animal symbols in their lunar calendar. Moreover, even among those American parents who are preoccupied with the Western zodiac, most would find it quite challenging to time their births in a particular month of the zodiac.

Will the lunar zodiac signs of Chinese children lead to differential treatment by families, friends, or society? One investigation suggests that they will not. Eberhard (1970), for instance, utilizing marriage data from China and Taiwan between the 17th and early 20th centuries, found no evidence that zodiacal characteristics affected marital chances. (Horoscopes of Chinese brides and grooms are often compared prior to marriage to determine their compatibility.) This author's preliminary investigations of contemporary Taiwanese marriage data support these findings. Similarly, in regard to the future value of zodiacal endowments, Singapore's current prime minister noted that the 13-member executive cabinet in early 1989 included three Pigs and two Dogs, but only one Dragon ("Forget the Zodiac Sign," 1989).

As far as lunar birth timing itself is concerned, I can only speculate whether it will continue for the duration of our postindustrial existence. To the extent that parental hopes and fears for their children continue to spring eternal, so may this unique initial investment in child quality. No matter what the future holds, we will be able to moni-

tor the manifestation of these preferences as long as vital registration continues. Patience will be required, however, because the animal zodiac proceeds in real time. Like comets, earthquakes, and identical twins, we cannot control the rhyme or reason from whence this symbolic natural experiment comes. It simply appears, bearing its own special logic that reminds us of two key principles: that cultural norms (strictly defined) can influence childbearing and that these norms, themselves, may emerge from modernity, rather than be eliminated by it.

NOTE

An earlier version of this article was presented at the 1996 annual meeting of the Association for Asian Studies in Honolulu, Hawaii. In Singapore, Paul Cheung at the Population Planning Unit of the Ministry of Health provided helpful advice as well as access to some of the newspaper clippings that are referred to in this article. A much earlier draft of this manuscript profited from suggestions by Jane Menken, S. Philip Morgan, Susan Naquin, Herbert Smith, and Susan Watkins. The author's field research throughout this project has been supported by the Andrew W. Mellon Foundation.

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